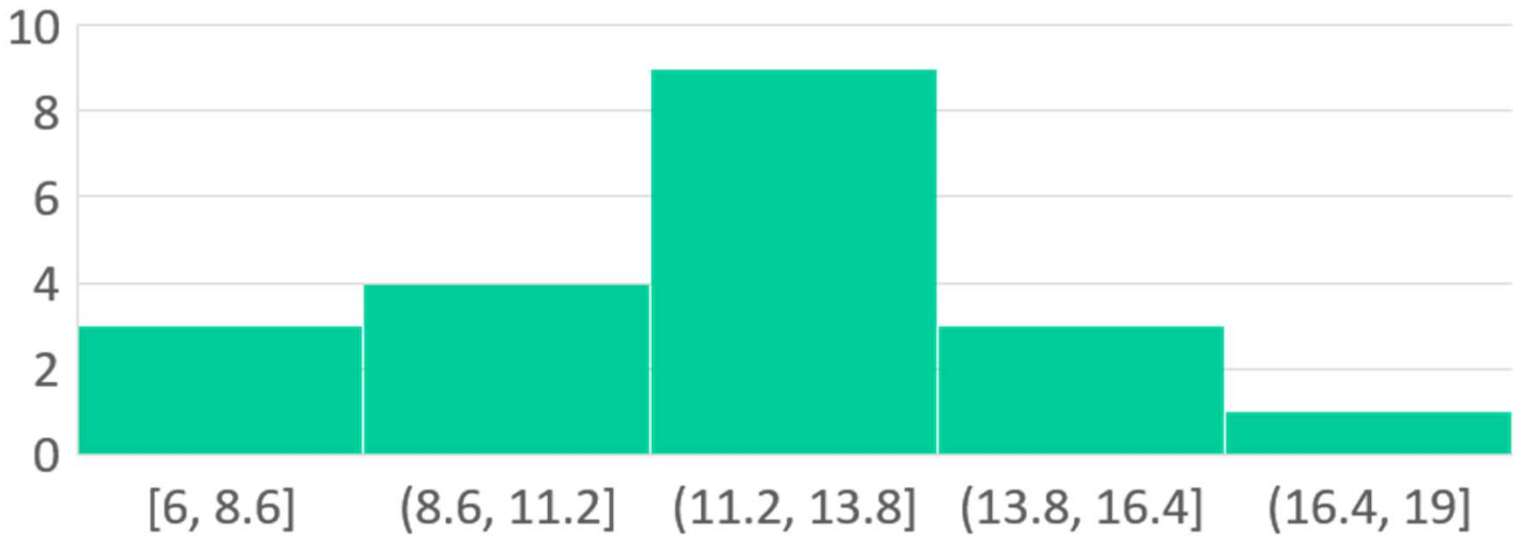
Central limit theorem

Given population and its distribution

19, 13, 15, 12, 13, 14, 13, 10, 13, 11, 15, 12, 13, 9, 7, 6, 7, 12, 13, 10

Mean = 11.85



Given population – sample mean

19, 13, 15, 12, 13, 14, 13, 10, 13, 11, 15, 12, 13, 9, 7, 6, 7, 12, 13, 10

Randomly select 5 points
Say 19, 12, 13, 7, 13
Compute mean = 12.8

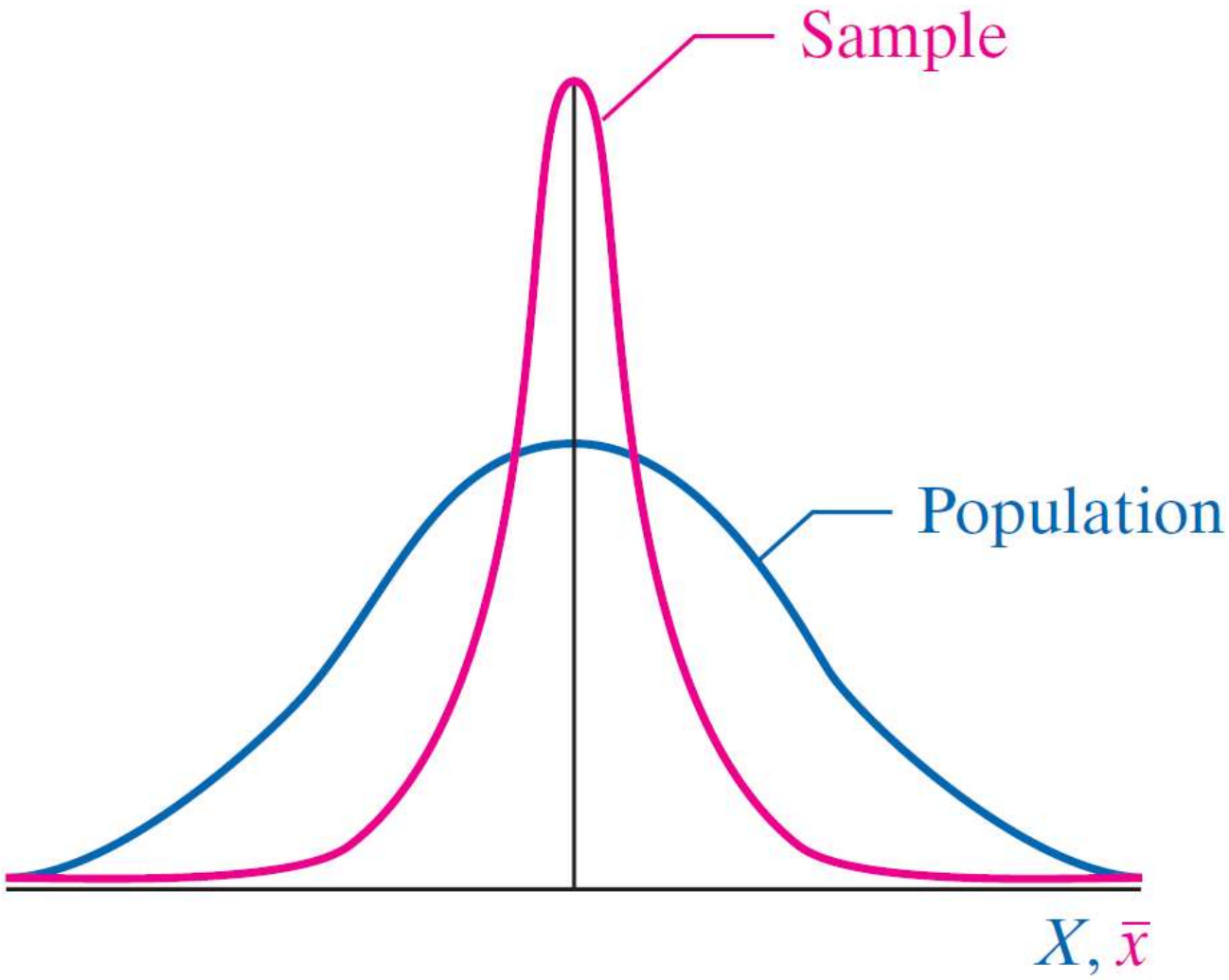
Repeat it many times
We get many means
Plot the distribution of the means

Central limit theorem

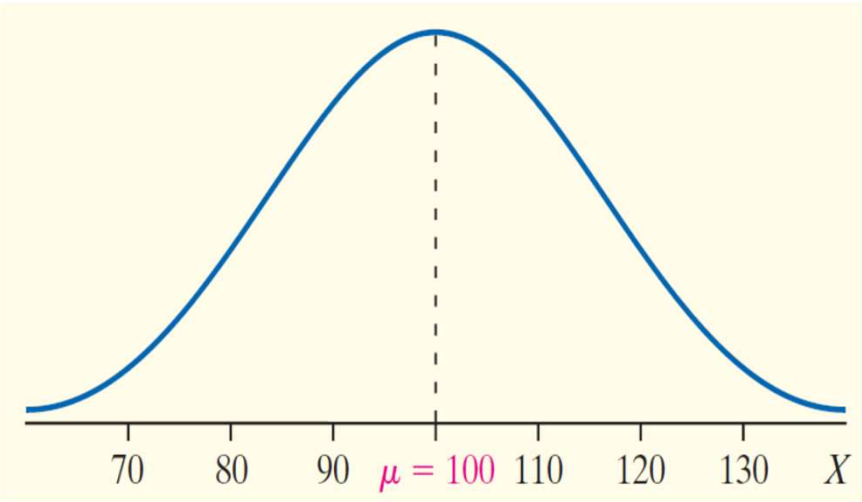
- n: Population size
- Distribution of n may be Gaussian or uniform or anything else

- Take samples
- Compute mean($\bar{x}$)
- Distribution of $\bar{x}$ is approximately normal
- Distribution shape original population does not matter

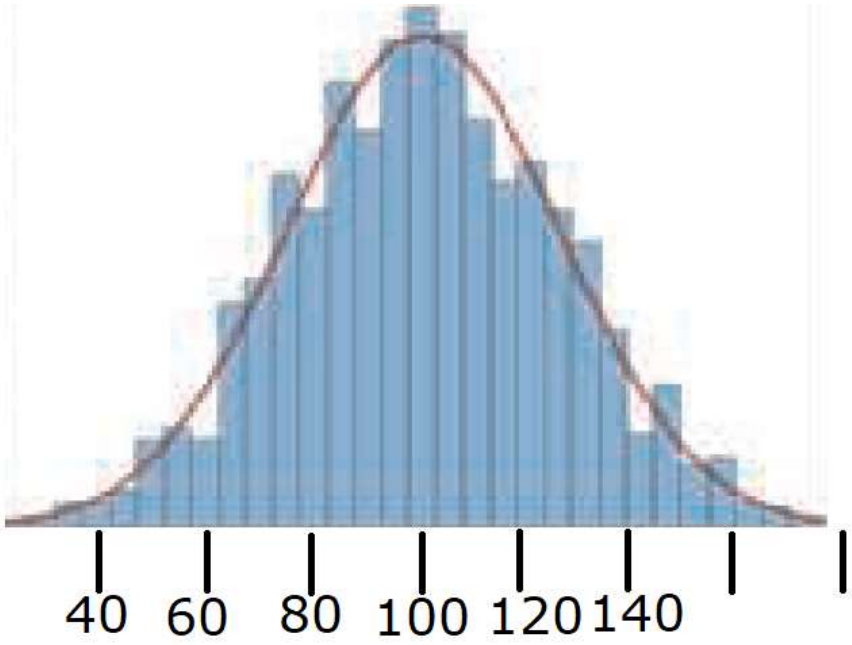
Distribution of original population and distribution of sample mean



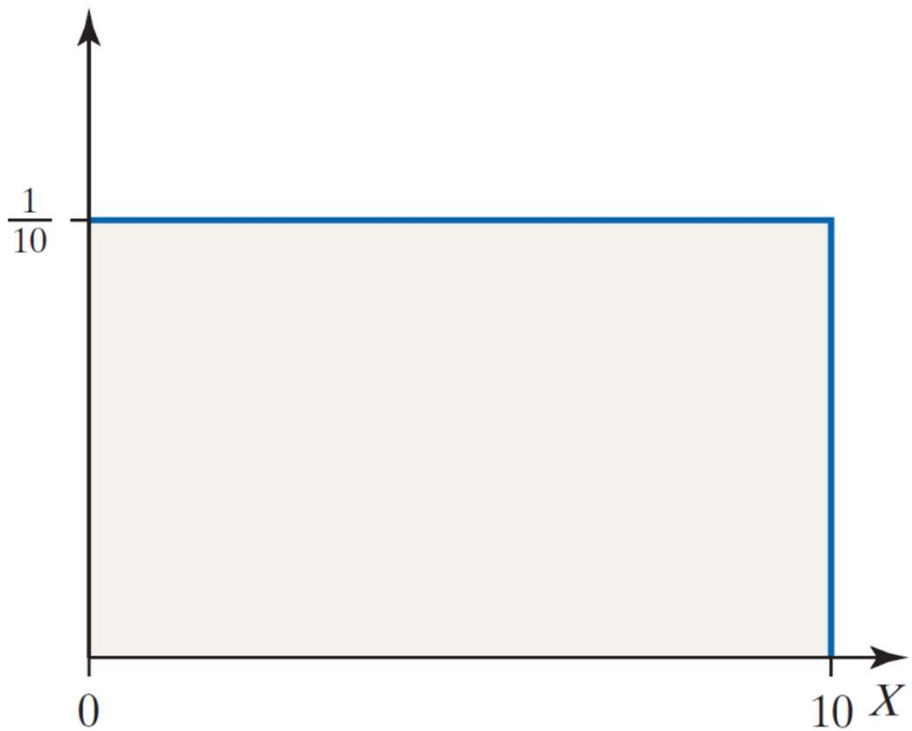
Original distribution is normal



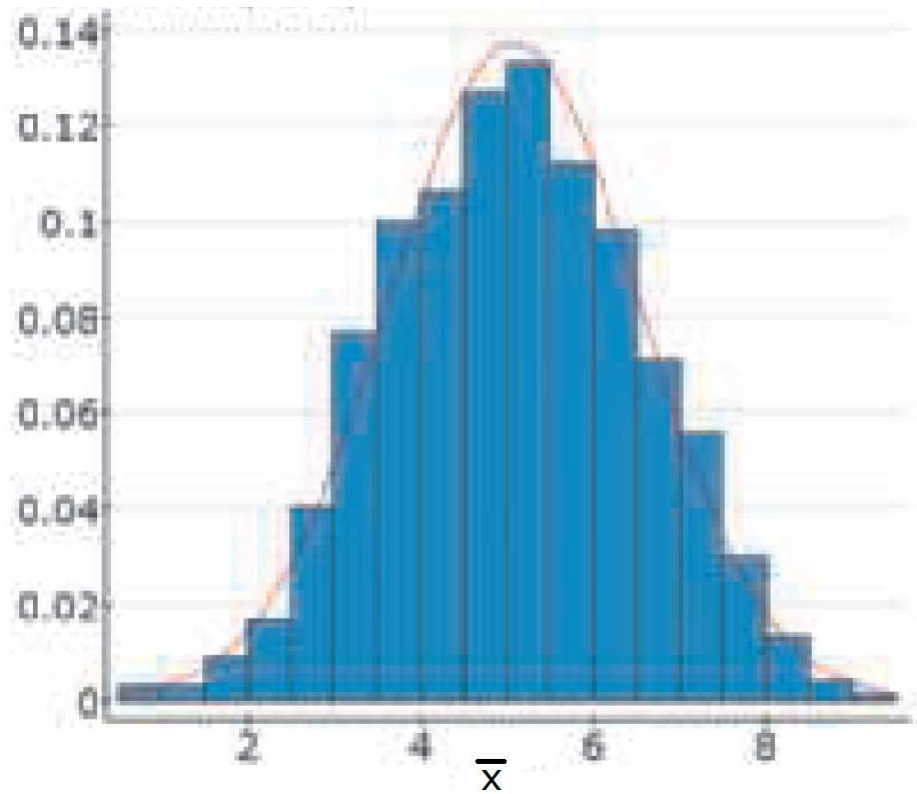
Distribution of sample means



Original distribution is uniform



Distribution of sample means



Normal distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \triangleq \mathcal{N}(\mu, \sigma)$$

Central limit theorem

$$\bar{X} = \mathcal{N}(\mu_{\bar{x}}, \sigma_{\bar{x}})$$

Distribution of sample means

n – sample size

Mean of "distribution of sample means":

$$\mu_{\bar{x}} = \mu_{original_population}$$

Variance of "distribution of sample means":

$$\sigma_{\bar{x}} = \frac{\sigma_{original_population}}{\sqrt{n}}$$